Red Team Recon & OSINT: Attack Surface Lookup Bot Using Shodan InternetDB

ABSTRACT

The project titled **"Red Team Recon & OSINT: Attack Surface Lookup Bot Using Shodan InternetDB"** focuses on automating basic reconnaissance and Open-Source Intelligence (OSINT) activities through a Telegram bot developed using n8n. The main aim of this project is to simplify the process of retrieving publicly available information about an IP address and presenting the results in a clear and structured format. This helps users understand basic attack surface information without manually searching multiple sources.

The system uses n8n to create an automated workflow that receives IP address queries through Telegram. When a user sends a command in the format /ip <IP address>, the workflow validates the input and checks whether the provided address is suitable for the lookup. If the input is valid, an HTTP Request node communicates with the **Shodan InternetDB API** to retrieve available information associated with the public IP address. The retrieved data may include open ports, hostnames, and Common Vulnerabilities and Exposures (CVE) references, depending on the information available for that address.

The workflow also includes a JavaScript-based rule that compares the discovered ports against a predefined watchlist containing ports 135, 143, and 443. Based on this custom rule, the system assigns a HIGH or NORMAL label. The AI Agent and Structured Output Parser are intended to organize the retrieved information into a readable report, while the message formatting and redaction steps help prepare the output before it is sent back to the user through Telegram.

The project supports testing with valid public IP addresses, invalid inputs, and private IP addresses. These checks help demonstrate input validation and the handling of different request conditions. The automated response can provide a summary of available network exposure information, matching watchlist ports, CVE references, and basic security recommendations where supported by the retrieved data.

This project demonstrates how workflow automation, OSINT techniques, API integration, and messaging platforms can be combined to support preliminary cybersecurity reconnaissance. However, the results depend on the data available through InternetDB and may not represent the current or complete state of a

target system. The HIGH or NORMAL label is based on a simplified custom rule and does not independently confirm that a vulnerability exists. Therefore, the system is intended for preliminary information gathering and educational purposes, with testing restricted to authorized and legitimate security activities.



